Hodgkin lymphoma

Classic HL vs NLPHL · highly curable, and therapy adapted by **interim PET** (omit **bleomycin and radiation** when PET-negative) · **N-AVD (US) and BrECADD (Europe)** are the advanced-stage standards

Compiled from the Stanford Oncology Handbook 2025, Lymphoma: Hodgkin (Chou L, Schroers-Martin J) · NCCN · primary trial reports verified against PubMed, September 2026.

CLASSIC HL VS NLPHL

	Classic HL	NLPHL
Tumor cell	**Hodgkin Reed-Sternberg (HRS)** — binucleate, owl-eye nucleoli; a tiny minority of cells in an inflammatory background	**LP ("popcorn") cells** in nodules of follicular dendritic cells, ringed by PD-1+ T-cell rosettes
Phenotype	**CD30+ CD15+, CD45–, PAX5 dim**, MUM1+. CD20 negative or weak/patchy; OCT2/BOB1 lost. EBV (LMP1) + in a subset	**CD20+ CD45+**, PAX5+, BCL6+, OCT2/BOB1 +. **CD30– CD15–**
Biology	**9p24.1 gain → PD-L1/PD-L2** + JAK-STAT, NF-κB — the reason PD-1 blockade works	Intact germinal-center B-cell program — really a B-cell lymphoma
Subtypes	Nodular sclerosis (young, mediastinal) · mixed cellularity (older, EBV+, HIV) · lymphocyte-rich · lymphocyte-depleted	**ICC 2022: nodular lymphocyte-predominant B-cell lymphoma**; WHO-HAEM5 keeps NLPHL. Variant patterns C–F (THRLBCL-like) do worse
Targets	**Brentuximab (CD30), anti-PD-1**	**Rituximab (CD20)** — no CD30 target

Lymphocyte-rich cHL mimics NLPHL on morphology — the immunophenotype separates them. Gray-zone lymphoma sits between cHL and PMBL.

WORKUP

- History: **B symptoms, pruritus, alcohol-induced nodal pain**; exam of nodal regions, liver, spleen; performance status
- **Excisional biopsy preferred**, core acceptable, **never FNA** (HRS cells are sparse and architecture matters)
- IHC — the critical panel: **CD30, CD15, PAX5, CD20, CD3**
- Labs: CBC/diff, CMP (**albumin**), **ESR**, LDH, **HIV**, HBV and HCV, pregnancy test
- **PET-CT** stages; **no routine marrow biopsy** — only for focal bone lesions or unexplained cytopenias
- **Echo/MUGA** before anthracycline · **PFTs with DLCO** before bleomycin · before PD-1: TSH, ACTH/cortisol, troponin
- **Fertility:** low risk with A(B)VD or N-AVD, so preservation is not mandated; **bank before eBEACOPP/BrECADD or salvage**

DEAUVILLE 5-POINT SCORE

1 · 2	No uptake · uptake **≤ mediastinum**
3	Uptake **> mediastinum but ≤ liver** — usually adequate response
4 · 5	Moderately **> liver** · markedly > liver or new lesions — **treatment failure**

Response is reported this way in practice and in every trial cited. (The handbook itself specifies only PET/CT at completion, plus marrow biopsy if positive at diagnosis.)

STAGING — COTSWOLDS-MODIFIED ANN ARBOR

I	**One** lymph node region or lymphoid structure (spleen, thymus, Waldeyer ring)
II	**≥2** regions, **same side** of the diaphragm. Hilar nodes are lateralized — bilateral involvement is stage II
III	Regions on **both sides** of the diaphragm
IV	**Disseminated** — marrow, CNS, or **more than one** extranodal site
A / B	B = weight loss >10% over 6 mo, **or** unexplained fever >38°C in the past month, **or** recurrent drenching night sweats
S · X · E	S splenic · **X bulky: node ≥10 cm**, or mediastinal mass ≥⅓ the transverse thoracic diameter at T5–6 · E limited direct extranodal extension

A single extranodal site as the only disease is IE; more than one extranodal site is stage IV. Hilar (bronchopulmonary) nodes sit outside the mediastinum.

EARLY-STAGE (CS I–II) UNFAVORABLE FACTORS

Factor	EORTC	GHSG	NCCN
Mediastinal mass	**>0.35** MTR	>⅓ CXR	>⅓ or >10 cm
Age ≥50	**yes**	—	—
ESR	≥50 (≥30 if B)	≥50 (≥30 if B)	≥50 **or any B**
Extranodal disease	—	**yes**	—
Nodal areas	**≥4** (EORTC areas)	**≥3** (GHSG areas)	>3

Favorable = none present. GHSG areas lump infraclavicular with ipsilateral cervical and both hila with the mediastinum. **Stage IIB with bulk or extranodal disease is treated as advanced** by GHSG (eligible for HD21). The handbook table lists ≥3 / ≥2 areas — the trial definitions are ≥4 (EORTC) and ≥3 (GHSG).

IPS — ADVANCED STAGE (FREEDOM FROM PROGRESSION)

Factors	Hasenclever 1998	Moccia 2012
0 · 1	84% · 77%	88% · 84%
2 · 3	67% · 60%	80% · 74%
4 · ≥5	**51% · 42%**	**67% · 62%**

One point each: **albumin <4 · Hgb <10.5 · male · stage IV · age ≥45 · WBC ≥15,000 · lymphocytes <600 or <8%**. Moccia is the ABVD-era revision. Modern regimens flatten the curve — N-AVD 3-yr PFS was 87% even at IPS 4–7.

TREATMENT PATHWAY

Early favorable (CS I–II, no risk factors)

ABVD ×2 → interim PET (PET2)
DS 1–2: + **20 Gy ISRT** (HD16, GHSG-favorable)
or ABVD ×1 + 30 Gy (H10F, RAPID)
no RT: **ABVD ×2 more** (H10F, CALGB 50604)
DS 3: ABVD ×2 + RT, or AVD ×4 (RATHL)
DS 4–5: escalate — eBEACOPP ×2 + ISRT (H10)

Early unfavorable

ABVD ×2 → PET2
DS 1–2: ABVD ×2 more + 30 Gy ISRT (H10U)
no RT: ABVD ×2 → **AVD ×4** if DS 1–3 (RATHL)
DS 4–5: eBEACOPP ×2 + 30 Gy
alt: nivo-AVD ×4 (NIVAHL) or BV-AVD (BREACH) + RT
GHSG 2+2: **omit RT if PET4-negative** (HD17)

Advanced (III–IV, GHSG IIB-high-risk)

Pick one backbone
N-AVD ×6 (S1826) — US standard, age ≥12
BrECADD ×4–6 (HD21) — age ≤60; 4 if PET2 neg
BV-AVD ×6 (ECHELON-1) — OS win vs ABVD
ABVD ×2 → PET2 → **AVD ×4** (RATHL)
>60 & fit: N-AVD; unfit: BV-DTIC, BV-nivo
EOT PET: RT only to avid residual (0.7% in S1826)

Relapsed / refractory

Salvage → PET-CR → HDT/ASCT
PD-1 combo first if checkpoint-naive:
pembro-GVD · pembro-ICE · nivo-ICE · BV-nivo
high risk → **BV ×16 consolidation** (AETHERA)

Relapse after ASCT / ineligible
pembrolizumab (KEYNOTE-204) > BV; nivolumab
allo-HCT for chemosensitive, fit patients

Hodgkin lymphoma

EARLY STAGE — OMITTING RT AFTER NEGATIVE PET

Trial	Design	Result
HD16 · H10 · RAPID *favorable / non-bulky*	ABVD ×2–3; PET-neg randomized ± RT	**Non-inferiority failed in all three** — PFS lower without RT (HD16 5-yr 93.4 vs 86.1%; RAPID 3-yr 94.6 vs 90.8%); **OS no different**
CALGB 50604 *non-bulky I–II*	ABVD ×2 → PET2-neg: ABVD ×2 more, **no RT**	3-yr PFS **91%** — the US chemo-only benchmark (single-arm)
HD17 *GHSG unfav*	eBEACOPP ×2 + ABVD ×2; PET4-neg: 30 Gy vs none	5-yr PFS 97.3 vs 95.1% — **NI met: omit RT** after intensive 2+2

The trade: dropping RT after a negative PET costs ~4–7 points of PFS with **no OS difference in any trial**. Lean to omission in young women with breast in the field, or when heart/coronaries are in the field; lean to RT for bulk, DS 3, or slow response.

BLEOMYCIN OMISSION

- **Pulmonary toxicity in 18%**, fatal in a quarter of those; 5-yr OS **63 vs 90%** with vs without it (Mayo, n=141). **Risk: age ≥40, ABVD, G-CSF** — omitting bleomycin did not reduce CR, PFS or OS
- **RATHL:** PET2-neg → AVD ×4 is non-inferior (3-yr PFS 84.4 vs 85.7%; confirmed at 7.3 yr) with less lung toxicity. PET2-pos → escalation, 3-yr PFS only 67.5%
- **Never with brentuximab** (contraindicated — pulmonary toxicity), and don't start it at all over 60 — ECHELON-1 ≥60: ABVD lung toxicity 13%, 3 deaths
- Stop permanently for cough/dyspnea + falling DLCO; **avoid G-CSF with ABVD**

ADVANCED STAGE — BACKBONE

Regimen	Evidence	Choose it / cost
N-AVD ×6	**S1826** vs BV-AVD (n=994, age ≥12): 3-yr PFS **91 vs 82%**, HR 0.48. 3-yr OS 98 vs 97%, **not significantly different — no OS benefit yet**. **FDA approved 20 Mar 2026**	**US default**, incl. adolescents and >60. Neuropathy 28 vs 54%; PROs favor it; G-CSF optional; RT in 0.7%. Benefit held in EBV+ (HR 0.30) and non-NS histology. Comparator was BV-AVD, not ABVD
BrECADD ×4–6	**HD21** vs eBEACOPP (n=1500, ≤60): treatment-related morbidity 42 vs 59%; 4-yr PFS **94.3 vs 90.9%** (HR 0.66); OS 98.6 vs 98.2%	**European standard** (EC-approved 2025; not FDA-labeled). Only 4 cycles if PET2-neg. Still intensive: G-CSF, PJP, febrile neutropenia. Gonadal recovery at 4 yr (men) **86 vs 40%** vs eBEACOPP
BV-AVD ×6	**ECHELON-1** vs ABVD (n=1334): 5-yr PFS 82.2 vs 75.3% (HR 0.68); **6-yr OS 93.9 vs 89.4%, HR 0.59** — the only OS win over ABVD	Now beaten by N-AVD head-to-head. Mandatory G-CSF, neuropathy. **No PFS benefit at ≥60** (5-yr 67 vs 62%) with grade 3–4 neuropathy 18%, FN 37%
eBEACOPP	Highest PFS of the chemo-only era (HD9/HD15/HD18)	**Legacy** — infertility (men 40% gonadal recovery at 4 yr), **t-MDS/AML 1.7%** after ≥4 cycles. Replaced by BrECADD
ABVD → AVD	**RATHL** PET-adapted (above); 5-yr ABVD ceiling ~75% PFS in ECHELON-1	Where novel agents are unavailable; PET2-pos patients do poorly

No head-to-head N-AVD vs BrECADD. Cross-trial comparison misleads: S1826 enrolled adolescents and >60 with investigator-read response; HD21 was ≤60 and included GHSG stage IIB with risk factors.

OLDER ADULTS (>60)

- **N-AVD if anthracycline-fit:** S1826 ≥60 (n=99) 2-yr PFS **89 vs 64%**, OS 96 vs 85% (subgroup); non-relapse mortality **6 vs 16%**; 6 cycles without dose reduction in 69 vs 26%
- **Avoid bleomycin** and be wary of BV-AVD (neuropathy, FN, no PFS gain at ≥60)
- **Sequential BV ×2 → AVD ×6 → BV ×4** (phase 2, n=48): 2-yr PFS 84%, but **45% if CIRS-G ≥10 and 25% with IADL loss** — do a geriatric assessment
- BrECADD at 61–75 (HD21 cohort, n=83): 2-yr PFS 91.5% but **FN 55%** — only for the very fit
- Anthracycline-unfit: **BV + dacarbazine** (CR 62%, mPFS 17.9 mo) · BV + nivolumab (CMR 48%) · **not BV + bendamustine** (stopped for toxicity)

RELAPSED / REFRACTORY

- **Rebiopsy** suspected relapse. Transplant-eligible: salvage → **HDT/ASCT**, which cures roughly half; **PET-negative going into ASCT is the key predictor**
- **PD-1 combinations first if checkpoint-naive** (all single-arm phase 2): pembro-GVD CR **95%** (n=39) · pembro-ICE CMR 86.5%, 2-yr PFS 87% · nivo → nivo-ICE 2-yr PFS 94% if bridged to ASCT · BV-nivo CR 67%, 3-yr PFS 91% with ASCT
- **BV consolidation ×16 after ASCT** for high risk (primary refractory, relapse <12 mo, extranodal at relapse): AETHERA 5-yr PFS **59 vs 41%** (HR 0.52), **no OS benefit**; less clear after BV- or PD-1-based salvage
- **After ASCT or ineligible:** pembrolizumab beat BV (KEYNOTE-204 mPFS **13.2 vs 8.3 mo**, HR 0.65; OS not reported). Single agents respond often but rarely completely: nivolumab ORR 69% (CheckMate 205), pembrolizumab ORR 69%, CR 22% (KEYNOTE-087); BV ORR 75%, CR 34%. Nivolumab R/R cHL converted to regular approval Mar 2026
- **Allo-HCT** for fit patients with chemosensitive relapse after ASCT. **Prior PD-1 raises early severe GVHD and non-infectious febrile syndromes** — plan with the transplant team

NLPHL (NLPBL)

- Indolent with **late relapses**; **transformation 14% at median 8.1 yr** (7% at 10 yr, 30% at 20 yr; **splenic involvement** predicts) — biopsy any progression
- **IA–IIA contiguous, non-bulky: ISRT 30–36 Gy**, or observe after complete excision
- I–IIA non-contiguous (GHSG-favorable): R-ABVD ×2 + 20 Gy · I–IIB or bulky: R-ABVD or R-CHOP ×4 + 30 Gy
- **III–IV: R-CHOP ×6** ± ISRT; rituximab alone or observation if asymptomatic, low burden, no variant pattern. Transformed/THRLBCL-like → treat as DLBCL

SURVIVORSHIP AND LATE EFFECTS

- **No routine surveillance PET** after CR — history, exam, CBC/CMP/ESR at visits
- **Breast cancer after chest RT:** cumulative incidence **35% by age 50** in childhood-HL survivors, mantle SIR 24. **Screen annually from 8–10 yr after RT or age 40, whichever first; add breast MRI if irradiated at age 10–30** (NCCN)
- **Second cancers:** SIR 4.6, 40-yr cumulative incidence 48.5% (Dutch cohort treated 1965–2000); axilla-sparing fields cut breast risk (HR 0.37). Lung (smokers), thyroid, GI, sarcoma in field
- **t-MDS/AML:** 0.9% overall, **1.7% after ≥4 eBEACOPP** (GHSG, n=11,952); median OS 7.2 mo
- **Cardiac:** anthracycline + mediastinal RT → CAD, valve disease, cardiomyopathy, pericardial disease — aggressive risk-factor control; consider stress test/echo at ~10-yr intervals; carotid US after neck RT
- **Thyroid:** annual TSH after neck/mediastinal RT (or PD-1). **Fertility:** low risk with A(B)VD/N-AVD, high with eBEACOPP

Hodgkin lymphoma

FRONTLINE DOSING

Regimen	Dosing (mg/m² unless stated)	Cycle
ABVD	Doxorubicin 25 · bleomycin 10 units/m² · vinblastine 6 · dacarbazine 375 — all d1 + d15. **No G-CSF**; don't delay for afebrile neutropenia	q28d
AVD	As ABVD without bleomycin (RATHL cycles 3–6)	q28d
N-AVD	**Nivolumab 240 mg** (3 mg/kg, max 240, if <18) + AVD — all d1 + d15 ×6. G-CSF optional	q28d
BV-AVD	**Brentuximab 1.2 mg/kg** (max 120 mg) + AVD — d1 + d15 ×6. **Primary G-CSF mandatory** (filgrastim d5–8 + d19–22)	q28d
BrECADD	**Brentuximab 1.8 mg/kg** (max 180 mg) d0 · etoposide 150 d1–3 · cyclophosphamide 1250 d1 · doxorubicin 40 d1 · dacarbazine 250 d2–3 · dexamethasone 40 mg d1–4. **No vincristine, no bleomycin**. G-CSF, PJP + antiviral prophylaxis; start dose level 4, de-escalate for toxicity	q21d ×4–6
eBEACOPP	Etoposide 200 d1–3 · doxorubicin 35 d1 · cyclophosphamide 1250 d1 · bleomycin 10 d8 · vincristine 1.4 (max 2 mg) d8 · procarbazine 100 d1–7 · prednisone 40 d1–14. G-CSF + PJP	q21d

Cumulative doxorubicin: ABVD/N-AVD/BV-AVD ×6 = 300 mg/m²; BrECADD ×6 = 240; eBEACOPP ×6 = 210. The handbook's BrECADD table lists vincristine and BV in mg/m² — both are errors (verified against Eichenauer, Lancet Oncol 2017).

TOXICITY AND MONITORING

- **BV neuropathy** (sensory, cumulative; AETHERA 56 vs 16%). Monotherapy: grade 2–3 → hold to ≤1, resume 1.2 mg/kg; grade 4 stop. With AVD: grade 2 → 0.9 mg/kg; grade 3 hold to ≤2 then 0.9. Boxed PML; pancreatitis
- **Bleomycin:** baseline PFTs/DLCO, repeat for symptoms; no G-CSF; warn anesthesia about high FiO_2
- **PJP (TMP-SMX) + acyclovir** with BrECADD, eBEACOPP and every BV-containing regimen
- **Checkpoint irAEs:** TSH at baseline and every cycle — thyroid dysfunction 7% (adolescent N-AVD), hypothyroidism 15% (NIVAHL); pneumonitis more with pembrolizumab than BV (KEYNOTE-204); adrenal, hepatitis, colitis → prednisone 1–2 mg/kg
- **Anthracycline:** echo at baseline; **dacarbazine** highly emetogenic; **vinblastine** vesicant — constipation, neuropathy

SALVAGE AND POST-ASCT DOSING

Regimen	Dosing	Cycle
Pembro-GVD	Pembrolizumab 200 mg d1; gemcitabine 1000 (800 post-HSCT) + vinorelbine 15 + liposomal doxorubicin 10 — d1 + d8	q21d ×2–4
ICE ± PD-1	Ifosfamide 5000 + mesna 5000 CIV d2 · carboplatin AUC 5 (max 800 mg) d2 · etoposide 100 d1–3. + pembrolizumab 200 mg d1, or nivolumab 240 mg d1 (after nivo alone q2w up to 6 doses)	q21d ×2 → collect
BV + nivolumab	Brentuximab 1.8 mg/kg (max 180) d1; nivolumab 240 mg C1D8, then d1 of C2–4. PJP + antiviral	q21d ×4
BV ± bendamustine	Brentuximab 1.8 mg/kg d1 ± bendamustine 90 d1–2 (salvage only — toxic in older frontline)	q21d ×6
BV consolidation	1.8 mg/kg (max 180), start 30–45 days post-ASCT	q21d ×16
PD-1 alone	Pembrolizumab 200 mg q3w or 400 mg q6w · nivolumab 240 mg q2w or 480 mg q4w	until PD

ICE: filgrastim from d8 or pegfilgrastim; hydration with bicarbonate. GVD d8: cut gemcitabine/vinorelbine 25% for ANC 500–1199 or plt 75–99K.

SPECIAL POPULATIONS AND EMERGING TOOLS

- **Adolescents 12–17** (24% of S1826): N-AVD 3-yr PFS **93 vs 82%** (HR 0.37), residual-site RT in 3 patients — one harmonized pediatric/adult protocol
- **HIV-associated cHL:** full-dose curative therapy on ART. S1826 enrolled CD4 ≥200 with undetectable viral load (n=16): 2-yr PFS 77.9% N-AVD vs 75.0% BV-AVD — too few to compare
- **EBV+ and non-nodular-sclerosis histology** historically do worse; N-AVD largely erased the gap (3-yr PFS EBV+ **91 vs 70%**, HR 0.30; non-NS 86 vs 63%)
- **ctDNA MRD after 2 cycles** (HD21 case-cohort): MRD-positive 4-yr PFS **36.7 vs 82.2%** (HR 5.3); adds to PET2 but is still investigational
- **PET pitfalls:** thymic rebound, brown fat, G-CSF marrow uptake, and sarcoid-like reactions on PD-1 — **biopsy a DS 4–5 residual before calling refractory disease**

KEY TRIALS

Trial	Setting / design	Result
HD16 · H10 · RAPID	Early stage; PET-neg after ABVD ± RT	RT omission: NI failed, OS equal
HD17 2021	GHSG unfav; 2+2, PET4-neg ± 30 Gy	5-yr PFS 97.3 vs 95.1% — **NI met**
CALGB 50604 2018	Non-bulky I–II; ABVD ×4 no RT if PET2-neg	3-yr PFS 91% (single-arm)
RATHL 2016	Advanced; PET2-neg ABVD vs AVD	3-yr PFS 85.7 vs 84.4% — NI (confirmed 7.3 yr)
ECHELON-1 2018	III–IV; BV-AVD vs ABVD (n=1334)	mPFS HR 0.77; **6-yr OS 93.9 vs 89.4%**, HR 0.59
SWOG S1826 2024	III–IV, age ≥12; N-AVD vs BV-AVD (n=994)	3-yr PFS **91 vs 82%**, HR 0.48; OS p=0.09
GHSG HD21 2024	Advanced, ≤60; BrECADD vs eBEACOPP (n=1500)	Morbidity 42 vs 59%; 4-yr PFS 94.3 vs 90.9%
AETHERA 2015	High-risk post-ASCT; BV ×16 vs placebo	mPFS 42.9 vs 24.1 mo; 5-yr 59 vs 41%; no OS gain
KEYNOTE-204 2021	R/R post-/non-ASCT; pembrolizumab vs BV	mPFS **13.2 vs 8.3 mo**, HR 0.65
KEYNOTE-087 · CheckMate 205	Single-arm PD-1 after ASCT and/or BV	ORR 69% both; CR 22% (KN-087)
BREACH · NIVAHL 2023	Early unfav, randomized phase 2	BV-AVD PET2-neg 82 vs 75%; nivo-AVD PFS ~100%
Pembro-GVD · BV-nivo	Single-arm first salvage	CR 95% / 67%; high post-ASCT PFS
Evens BV→AVD→BV 2018	Age ≥60, phase 2 (n=48)	2-yr PFS 84%; poor if CIRS-G ≥10